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January 9, 1968
SUPERSEDING
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FEDERAL SPECIFICATION

DISINFECTANT, GERMICIDAL AND FUNGICIDAL, CONCENTRATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of germicidal and fungicidal disinfectant concentrate suitable for general disinfection (see 3.11.2 and 6.1).

1.2 Classification.

1.2.1 Types. This specification covers disinfectant concentrates of the following types:

Type L - Liquid (water being the principal solvent).
Type S - Solid.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

L-P-391	-	Plastic, Sheets, Rods, and Tubes, Methacrylate, Cast.
GG-H-80	-	Handle and Blade Set, Surgical Knife Detachable Blade.
QQ-S-766	-	Steel Plates, Sheets and Strip - Corrosion Resisting.
ZZ-G-421	-	Gloves, Surgeon's.
PPP-B-636	-	Box, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-P-704	-	Pails, Shipping, Steel (1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-B-26701 - Bottles, Screwcap and Carboys, Poleythylene, Flexible, Plastic.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Insecticide, Fungicide, and Rodenticide Act.

Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act (Ch. III, Pt. 362 of the Code of Federal Regulations.)

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Research Service, Pesticide Regulation Division, Washington, D. C. 20250.)

U. S. Pharmacopeial Convention Publication, Inc:

Pharmacopeia of the United States.

(The Pharmacopeia of the United States is distributed by the Mack Publishing Company, Easton, Pennsylvania.)

(Technical societies and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified at the time of submission of a bid, the bidder shall submit to the contracting officer a certified copy of a recent laboratory report covering the quality and grade of materials which he proposes to deliver. The laboratory report shall contain test data which demonstrate that the disinfectant concentrate which the supplier proposes to deliver has been tested in accordance with the tests in section 4 and found to comply with the requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(a) An independent or commercial laboratory report.
(b) The prospective supplier's own laboratory report.
(c) A Governmental laboratory report from a contract.

The purpose of the above requirement is to assist the Government to determine the capability of bidders to conform to all the requirements of this specification. The submission of an acceptable report under this requirement shall not be construed as relieving a supplier from subsequently meeting all requirements of this specification on all deliveries.

3.1.1 Certificates. The supplier shall submit to the contracting officer or his representative, certificates certifying that the material he proposes to deliver does not contain the ingredients specified in 3.2.1 and that the material conforms to the storage stability requirement specified in 3.9.

3.2 Materials. The materials used in the disinfectant concentrate shall be of such physical characteristics and in such proportions as to conform to the requirements of this specification.

3.2.1 Excluded ingredients. The disinfectant concentrate shall contain no mercury compounds, peroxides, formaldehyde, or materials which release such compounds when diluted according to directions for the item.

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3.3 Antimicrobial requirements (bactericidal and fungicidal action). The disinfectant shall kill 99.99 percent (bacterial species) and 99.90 percent (fungal species) of the test organisms specified in table I within a 10-minute exposure period at 20° C. (68° F.) when diluted in distilled water and tested as specified in 4.3.6.

TABLE I. Bactericidal and fungicidal requirements

Organism	Effective dilution (see 4.4.1)	
	In the absence of horse serum	In the presence of horse serum
Micrococcus pyogenes var. aureus 209 (ATCC No. 6538)	1:100	1:50
Salmonella schottmuelleri (ATCC No. 9282)	1:100	1:50
Trichophyton interdigitale (ATCC No. 9533)	1:100	1:50

3.4 Acute oral toxicity. When tested as specified in 4.3.6, the disinfectant shall not kill more than five of a group of 10 rats (LD_{50} (oral) = 4 ml./kg. of body weight).

3.5 Dermatitis. The disinfectant concentrate shall cause no dermatitis when diluted and tested as specified in 4.3.6.

3.6 Effect on plastics and rubber. A 1:50 dilution of the disinfectant concentrate in distilled water shall cause no stickiness, no change in color, or no crazing when tested as specified in 4.3.6.

3.7 Odor. A 1:50 dilution of disinfectant concentrate in distilled water shall be substantially odorless when tested as specified in 4.3.6.

3.8 Solubility and emulsion stability. A 1:50 dilution of disinfectant concentrate in distilled water shall form a practically clear solution or homogeneous emulsion. If an emulsion is formed, the 1:50 dilution shall show no evidence of unemulsified oils nor separation of the emulsion when tested as specified in 4.3.6.

3.9 Storage stability. The disinfectant concentrate shall have no loss of antimicrobial potency after 1 year storage. Type L concentrate shall not form an amount of sediment exceeding 5 percent of the total volume of the liquid when tested as specified in 4.3.6.

3.10 Corrosive effect. A 1:50 dilution of disinfectant concentrate shall show no corrosive effect on stainless steel and not more than an average of 12.0 milligrams loss in weight of galvanized iron when tested as specified in 4.3.6.

3.11 Labeling.

3.11.1 Labeling. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act and in compliance with its requirements. Appropriate warning or caution statements required by the Act or required to insure safe usage of the material shall be included.

3.11.2 Unit containers. Each container of disinfectant concentrate shall be clearly and legibly lithographed, stenciled, or printed with a noncorrosive weather-resistant ink which will not smear nor rub off during handling. If all of the label data cannot be applied due to its being too voluminous, that portion, such as the mixing instructions and directions for use may be placed on a label made of white paper as specified in MIL-STD-129. Labels shall be securely affixed in place with water-resistant label adhesive as specified in section 3 of MIL-STD-129 and shall be waterproofed by coating the entire outer surface of the label with the same adhesive. The label data shall be as follows:

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DISINFECTANT, GERMICIDAL AND FUNGICIDAL, CONCENTRATE

PERCENT BY WEIGHT

ACTIVE INGREDIENTS.
(Name of ingredients)

INERT INGREDIENTS.
(Name of ingredients)

Total _____
100 percent

Specification
Stock number
Type
Manufacturer's designation
Manufacturer's name and address
Date of pack
Lot number
Contract number
Net contents

ALWAYS DILUTE ACCORDING TO "DILUTION TABLE" BEFORE USING

DIRECTIONS FOR USE

(For type S concentrates of multiple strength, appropriate modifications shall be made.)

DO NOT ADD TO SOAP OR DETERGENT SOLUTION

1. For routine use. Swab, wipe, or rinse surfaces (such as floors, walls, urinals, washstands, grabage pails), which have been previously cleaned with a soap or detergent solution and rinsed thoroughly, using 1 percent solution (see " DILUTION TABLE").

2. For special use.

A. If known heavy contamination exists, use 2 percent solution on surfaces. Swab, wipe, or rinse surfaces which have been previously cleaned with a soap or detergent solution and rinsed, with disinfectant solution at least twice, allowing solution to act for at least 10 minutes between applications. Disinfect smaller objects by immersion for at least 10 minutes. Permit all surfaces to dry before use.

B. Cold disinfection of surgical instruments. Sterilization by autoclave shall be performed wherever possible. Cold disinfection of surgical instruments with this material may be used as an emergency alternate. In the absence of heat, immerse in 2 percent solution for at least 10 minutes after removing excess soil by cleaning.

3. Precaution. To disinfect RUBBER and PLASTIC objects, swab surfaces if possible. Immerse only if required. LIMIT EXPOSURE TO 10 MINUTES. PROLONGED IMMERSION IN DISINFECTANT MAY AFFECT PLASTICS AND RUBBER SERIOUSLY.

WARNING

HARMFUL IF SWALLOWED

Do not get in eyes or on skin

In case of contact with skin, wash thoroughly with soap and water; for eyes, flush with plenty of water for at least 15 minutes and get medical attention.

ANTIDOTE

(Supplier shall furnish antidote for specific disinfectant.)

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DILUTION TABLE

Dilution	Ounce concentrate per gallon water	Ml. concentrate per liter water	Teaspoons concentrate per gallon water
1 percent solution (Routine use)	1-1/4	10	10
2 percent solution (Special use)	2-1/2	20	20

3.12 Average net content. The average net content per unit container shall not be less than specified.

3.13 Workmanship. The disinfectant concentrate shall be clean, uniform, and free from foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates. Where certificates are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.5. The lot size shall be expressed in units of filled unit containers of the specified capacity for the examinations in 4.3.2 and 4.3.3 and in units of shipping containers for the examination in 4.3.4.

4.3.2 Examination of the filled unit container. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container	Not in accordance with contract requirements. Not type, size, and class specified. Not amber colored glass (as applicable). Outer and inner seals not as specifed. Exterior coating not as specified. Closure not as specified. Any evidence of seepage or leakage from any part of the container including closure and seams. Bulged or distorted container.
Contents	Not type specified. Not clean and uniform. Not free from foreign matter.
Marking	Not as specified, missing, incomplete, incorrect, illegible or smeared. Unit container not labeled as specified (see 3.11).

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4.3.3 Examination for average net content. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Average net content	The average net content per container is less than specified.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking requirements comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. The lot shall be the number of shipping containers submitted for inspection at one time. Closed shipping containers shall be examined for closure defects.

Examine	Defect
Markings	Incorrect; illegible; omitted; of improper size, location, sequence or method of application (see 5.3).
Materials	Any nonconforming component, component missing, damaged, or otherwise defective.
Workmanship	Incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping, or stapling, as applicable. Bulged or distorted container.
Containers	Not as specified.
Contents	Quantity in shipping container not as specified.

4.3.5 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size and acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.2	I	2.5
4.3.3	S-2	N/A
4.3.4	S-2	2.5

4.3.6 Testing of the end item. The methods of testing listed in table II shall be followed for each lot. The lot size shall be expressed in gallons (type L) or pounds (type S) of disinfectant concentrate prepared from one batch manufactured in a single production and offered for inspection at one time. The sample unit for testing shall be 1-pint samples (type L) or 1-pound samples (type S) selected at random throughout the lot. The sample size shall be as indicated in the table below. All tests shall be performed on the composite sample as specified in table II. The composite sample shall be 2-pints or 32 ounces of disinfectant concentrate, selected in equal portions from each sample unit. The composite shall be thoroughly mixed, yielding a homogeneous mixture. Each sample unit and the composite shall be placed in separate, sealed, airtight and watertight metal or glass containers with identifying lot and date. Care shall be exercised to prevent the loss or absorption of moisture or loss of volatile material during the sampling, weighing, and storage of the specimens and samples. The lot shall be unacceptable if one or more sample units or the composite sample fail to meet any test requirement specified.

Lot size (gallons or pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. Directions for testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit composite	Results reported as		Inspection level	AQL
	Requirement	Test method	Individual unit	Lot average		Pass or fail	Numerically to nearest		
Excluded ingredients	3.2.1	1/							
Antimicrobial requirements	3.3	4.4.2		X	2		Average of 2 to 0.001 percent		
Acute oral toxicity	3.4	4.4.3		X	1	X			
Dermatitis	3.5	4.4.4		X	1	X			
Effect on plastics and rubber	3.6	4.4.5		X	1	X			
Odor	3.7	4.4.6		X	1	X			
Solubility and emulsion stability	3.8	4.4.7		X	1	X			
Storage stability	3.9 and 3.1.1	4.4.8		X	1	X			
Corrosive effect	3.10	4.4.9		X	1	X			

1/The supplier shall furnish to the contracting officer or his representative a certificate of material covering this requirement (see 3.1.1).

4.4 Test methods.

4.4.1 Preparation of test dilutions. One percent (1:100) and 2 percent (1:50) dilutions of the disinfectant concentrate shall be prepared in sterile distilled water and kept at 20° C. for at least 30 minutes before tests are begun. For type L concentrates, dilutions shall be volume/volume (ml./100 ml.). For type S concentrates, dilutions shall be weight/volume (gm./100 ml.). If type S concentrate is purchase (see 6.2), as being a multiple of the strength specified in 3.3, there shall be a preliminary dilution to the standard strength defined therein.

4.4.2 Antimcrobial requirements (bactericidal and fungicidal action).

4.4.2.1 Propagation of test organisms. Stock cultures of Micrococcus pyogenes var. aureus 209 (ATCC No. 6538) and Salmonella schottmuelleri (ATCC No. 9282) shall be maintained on nutrient agar consisting of 1.0 percent peptone, 0.5 percent beef extract and 0.5 percent NaCl (pH 7.0) at 20° C. New stock cultures shall be made each month by transferring a loopful of growth from a previous stock culture, incubating this new culture at 37° C. for 24 hours, and then storing at 20° C. For the test proper, nutrient broth of the same formula specified herein shall be employed in preparing test culture. Growth from a stock culture shall be transferred to a tube containing 10 ml. of nutrient broth and the tube shall be incubated at 37° C. for 24 hours. Serial transfers shall be made into nutrient broth on successive days after incubating the cultures at 37° C. for 24 hours in each instance. After the fourth consecutive daily transfer, cultures may be used in preparing the inoculum specified in 4.4.2.2. A new sequence of successive broth cultures shall be initiated from agar twice a month and no broth culture shall be used for the preparation of the inoculum specified in 4.4.2.2 if the number of serial transfers has been greater than twenty. Both bacterial species shall be handled in an identical fashion. For the preparation of stock cultures of the fungus, Trichophyton interdigitale (ATCC No. 9533) shall be grown for 12 to 15 days at 28° to 30° C., on a medium consisting of 0.2 percent tryptose, 0.1 percent yeast extract, 0.1 percent beef extract, 1.0 percent glucose and 2.0 percent agar, pH 7.2 and subsequently refrigerated at 5° C. To reduce degeneration, two primary stock cultures shall be prepared, overlaid with sterile mineral oil, and refrigerated at 5° C. until used. When secondary stock cultures are required, a set of six cultures shall be prepared from one of the primary stock cultures using the same medium and growth conditions specified for primary stock cultures. All secondary stock cultures not used immediately shall be refrigerated without mineral oil overlay after incubation. Secondary stock cultures shall be used to inoculate liquid medium for the production of the spore suspension. To prepare fungal spores for use in the inoculum specified in 4.4.2.2, 8.0 ml. of water shall be added to a secondary stock culture, the spores scraped off the slant with a stiff wire, the tube shaken vigorously and 1.0 ml. of this spore suspension added to a bottle of liquid medium identical with the stock fungus agar but without agar. The medium shall be dispensed in 12 ml. quantities in 4 ounce square bottle containing sufficient glass beads to just break the surface of the liquid when the indicated quantity of medium is employed. These seeded bottles shall be incubated at 28° to 30° C. for 12 to 15 days,

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then shaken thoroughly, and the contents filtered through sterile absorbent cotton. A microscopic spore count shall be performed on the filtrate using a hemocytometer counting chamber, and the spore count diluted to 8,000,000 conidia/ml. with sterile broth. This suspension shall be used in preparing the inoculum specified in 4.4.2.2. Liquid spore cultures may be prepared by serial subculturing from previously grown liquid medium but the number of successive subcultures shall not exceed three before returning to a new tube containing a secondary stock culture. Spore suspensions prepared and diluted to 8,000,000 conidia/ml. may be used 1 week after preparation if they have been refrigerated at 5° C. while not in use. The phenol resistance of each species of test organism shall be determined independently as follows: Appropriate dilutions of phenol in distilled water shall be prepared and dispensed in 5.0 ml. quantities in test tubes. The phenol used shall meet the requirement of the United States Pharmacopeia and shall have a congealing point above 40° C. (104° F.). Tubes containing phenol dilutions shall be placed in a water bath at 20° C. until the contents assume this temperature. Five-tenths ml. of each test culture shall be added to each tube of appropriate phenol dilution from a pipette, the time being noted, and each tube shall be immediately agitated. The culture shall be shaken 15 minutes before use and allowed to settle. Its temperature shall be 20° C. (68° F.) at the time of transfer. At the end of 5-, 10-, and 15-minute intervals, thoroughly mix the contents of each tube and transfer a loopful of inoculated disinfectant from each tube to a subculture tube containing 10 ml. of either nutrient broth (M. pyogenes var. aureus and S. schottmuelleri) or tryptose-yeast extract-beef extract-glucose broth (T. interdigitale). All nutrient broth subcultures shall be incubated at 37° C. for 48 hours. Tryptose-yeast extract-beef extract-glucose broth subcultures shall be incubated at 28° to 30° C. for 14 days. All tubes shall be observed for growth at the end of the respective incubation periods. The test cultures shall show the resistances to water dilutions of phenol as specified in table III.

TABLE III. Culture resistance

Culture	Dilution	Resistance minimum-minutes			Dilution	Resistance maximum-minutes		
		5	10	15		5	10	15
Micrococcus pyogenes var. aureus	1:70	---	---	---	1:70	+	---	---
Micrococcus pyogenes var. aureus	1:80	+	---	---	1:80	+	+	+
Micrococcus pyogenes var. aureus	1:90	+	+	---	N/A	N/A	N/A	N/A
Salmonella schottmuelleri	1:80	---	---	---	1:80	+	---	---
Salmonella schottmuelleri	1:90	+	---	---	1:90	+	+	+
Salmonella schottmuelleri	1:100	+	+	---	N/A	N/A	N/A	N/A
Trichophyton interdigitale	1:60	---	---	---	1:60	+	---	---
Trichophyton interdigitale	1:70	+	---	---	1:70	+	+	+
Trichophyton interdigitale	1:80	+	+	---	N/A	N/A	N/A	N/A

Note: + indicates growth; --- indicates no growth; N/A indicates not applicable.

4.4.2.2 Preparation of inoculum. The following mixtures shall be prepared and employed as inocula in the test:

Inoculum A

1.0 ml. of a 24-hour nutrient broth culture of Micrococcus pyogenes var. aureus.
1.0 ml. of a 24-hour nutrient broth culture of Salmonella schottmuelleri.
1.0 ml. of a spore suspension of Trichophyton interdigitale prepared as specified in 4.4.2.1.
0.6 ml. of normal horse serum.

Inoculum B

Same as inoculum A except for the substitution of sterile nutrient broth for the normal horse serum.

4.4.2.3 Test procedure. Carriers for the test shall consist of scrupulously clean corrosion resisting steel sheets, cut into squares 1 by 1 by 1/16 inch, conforming to class 304, condition "A", finish No. 4, 16 gage of QQ-S-766. Twenty squares shall be employed in the complete test, and the test may be run in two separate parts using 10 squares in each test. Ten squares shall be inoculated with 0.025 ml. of inoculum A specified in 4.4.2.2 and 10 squares with inoculum B. The inoculum on each square shall be spread to within 1/8 inch of each edge of the square and as uniformly as possible using a bent platinum wire. The inocula shall be dried on the squares in a 20° C. incubator having a relative humidity between 50 and 75 percent. Within 30 minutes after the squares have been dried, 0.04 ml. of water or disinfectant concentrate dilution shall be added to each square, the water or disinfectant shall be mixed intimately with the dried film of organisms, and the mixture spread to within 1/16 inch of each edge of the square. The water or disinfectant

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shall be permitted to act for exactly 10 minutes at 20° C. The square shall be inoculated according to the following pattern:

(a) Two control squares containing inoculum A shall be treated with 0.04 ml. of distilled water.
(b) Two control squares containing inoculum B shall be treated with 0.04 ml. of distilled water.
(c) Eight test squares containing inoculum A shall be disinfected with the 2 percent dilution of disinfectant concentrate.
(d) Eight test squares containing inoculum B shall be disinfected with the 1 percent dilution of disinfectant concentrate.

4.4.2.3.1 After the 10-minute exposure period, each square shall be placed in a separate round jar containing 20 ml. of diluent, the contaminated surface rubbed with a policeman, and the square shaken thoroughly in the liquid. The diluent shall consist of 0.406 percent, KH_2PO_4 and 0.655 percent, Na_2HPO_4 in distilled water and shall be supplemented by the addition of inhibitors when phenols, quaternary ammonium or halogen disinfectants are tested. For testing phenols and cresylics, distilled water may be used as a diluent or a nonionic detergent inhibitor of the appropriate concentration; for testing quaternary ammonium compounds, 0.07 percent Azolectin in aqueous 0.5 percent Tween 80 is the inhibitor; for testing halogen formulations, 0.5 percent sodium thiosulfate is employed as the inhibitor. One ml. of diluent from each jar containing test squares shall be plated out in each of three selective media; penicillin nutrient agar containing 0.3 units per ml. agar of penicillin G (potassium salt), polymyxin nutrient containing 150 units of polymyxin B sulfate per ml. agar, and terramycin nutrient agar containing 4 micrograms of terramycin hydrochloride per ml. agar. Appropriate dilutions of the diluent containing the four control squares shall be plated similarly in the three differential media. All plates containing penicillin nutrient agar and polymyxin nutrient agar shall be incubated at 37° C. for 48 hours. Terramycin agar plates shall be incubated at 28° to 30° C. for 5 days. After incubation, the number of colonies on each plate shall be counted and recorded. From this, the number of survivors of each species recovered from each square shall be calculated by multiplying the number of colonies per plate times the correct dilution factor. An average number of survivors per square of each species recovered from all eight test squares inoculated with inoculum A shall be calculated. Similarly, the average number of survivors per square of each species from the two control squares inoculated with inoculum A shall be calculated. The average percentage reduction for each species shall then be obtained from the following relationship and shall be rounded off to the second decimal place (bacteria) or first decimal place (fungus):

$$\text{Percent reduction of inoculum A} = \frac{\text{average number of survivors of a given species obtained on the eight test squares}}{\text{average number of survivors of a given species obtained on the two control squares}} \times 100$$

The percentage reduction obtained from the 10 squares inoculated with inoculum B shall be calculated in a similar manner. An example of the method of calculation is specified in table IV. Tests in which control squares yield average numbers of survivors less than the following minima for the indicated species shall not be acceptable: M. pyogenes var. aureus, 100,000 survivors per squares; S. schottmuelleri, 50,000 survivors per square; T. interdigitale, 10,000 survivors per square.

TABLE IV. Method of calculating results

Square number	Polymyxin agar				Penicillin agar				Terramycin agar			
	A	B	C	D	A	B	C	D	A	B	C	D
Test 1	1	20			0	0			0	0		
2	0	0			0	0			0	0		
3	3	60			0	0			0	0		
4	0	0	77.5		0	0			3	60		
5	3	60		99.99	0	0	2.5	99.99	1	20	10	99.99
6	2	40			0	0			0	0		
7	22	440			1	20			0	0		
8	0	0			0	0			0	0		
Control 1	---	1,400,000			---	250,000			---	10,000		
			1,160,000				275,000				15,000	
2	---	920,000			---		300,000		---	20,000		

A - Colonies per plate - number of colonies found on plates in which 1.0 ml. of diluent was planted.
B - Survivors per square.
C - Average number of survivors per square.
D - Average percentage reduction.

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4.4.3 Acute oral toxicity. Ten healthy albino rats of the Wister strain, weighing between 150 and 250 gm., shall be used for the oral toxicity test. The animals, none of whom shall be pregnant, shall be fasted for 24 hours preceding dosing. A wooden tongue depressor, narrowed at its central width and with a hole bored into its center large enough to admit a No. 10 French rubber catheter, shall be used as a gag to hold open the rat's mouth and permit entrance of the catheter. The test rats shall be weighed and a 1 ml. tuberculin syringe to which a No. 10 French catheter has been attached shall be filled with a measured amount of the disinfectant concentrate (4 ml. per kilogram of body weight). The rat shall then be held in one hand while the narrow edge of the center of the tongue depressor shall be held firmly with the other hand against the anterior teeth of the animal thus forcing the mouth open. Place the flat edge of the gag against the tongue. By turning the gag through a half-turn (counter-clockwise), the tongue is pulled forward and the bored hole in the depressor is approximately with the pharynx. With the animal's jaws open, the depressor is locked behind the canine teeth. The forefinger of the operator's hand holding the rat is then hooked over the end of the gag. With the other hand, insert the catheter firmly through the hole in the gag, throught the pharnyx and esophagus and into the stomach. When certain that the tip of the catheter is in the stomach, depress the plunger of the syringe to deliver the calculated amount of disinfectant. Withdraw catheter, remove gag, and place animal in normal environment with usual food and water. Keep the animals under observation for a period of 1 week.

4.4.4 Dermatitis. Five healthy, adult albino rabbits, weighing between 2 and 3.5 kg., shall be used. Both sides shall be closely clipped. (Hair shall not be removed by shaving or depilatories.) Pads of cotton gauze, approximately 1 inch square and two layers in thickness, shall be prepared. Pads shall be saturated with 1 ml. of the solutions shown in table V and shall be immediately fastened to the proper side of the animal with waterproof adhesive tape or equivalent means. The pads shall be completely covered in order to minimize evaporation. At the end of 8 hours, the pads shall be removed. The areas shall be observed at the end of 8, 24, and 72 hours from the start of the test. The disinfectant under test shall not have caused more irritation than the maximum of the other patch tests.

TABLE V. Patch tests

Pad No.	Solution	Position
1	Sterile distilled water	Right side
2	1:50 dilution in water of liquor cresolis saponatus, U.S.P. XII	Right side
3	1:50 dilution in water of the disinfectant concentrate	Left side

4.4.5 Effect on plastics and rubber. A 2-inch square of methacrylate plastic conforming to L-P-391 and a rubber glove conforming to grade A of ZZ-G-421 shall be completely immersed in a 1:50 dilution of disinfectant concentrate for 30 minutes at 20° to 30° C. (68° to 86° F.). A visual examination shall be made for stickiness, change in color and crazing of plastic after this exposure.

4.4.6 Odor. A 6-inch square of bleached cotton sheeting shall be immersed in a 1:50 dilution of disinfectant concentrate in distilled water, the sheeting shall be thoroughly wetted, wrung to remove excess liquid and hung up to dry for 48 hours in a well ventilated room. At the end of 48 hours, the treated section of sheeting shall be examined for residual odor.

4.4.7 Solubility and emulsion stability. Solubility and emulsion stability shall be determined as follows: Place 98 milliliters of distilled water at 25° ± 3° C. (77° ± 5.4° F.) in a 350 milliliter Griffin low form beaker (approximately 85 mm. high and 65 mm. diameter). Stir the water at 1000 r.p.m. using a stirrer consisting of a glass rod (10 mm. diameter and 50 mm. long) to which a 5 mm. diameter glass rod has been sealed at right angles to make a T-shaped stirrer. The stirrer should be positioned to clear the bottom of the beaker just enough to permit free rotation. Add 2.0 ml. of the disinfectant concentrate (type L) to the agitated water and continue stirring for 1 minute. For type S use 2 grams of concentrate. Pour the solution or emulsion thus formed into a 100 milliliter graduated, glass-stoppered cylinder, note the time and set aside for 24 hours at 25° ± 3° C. (77° ± 5.4° F.). Immediately after that period examine the solution or emulsion carefully for signs of unemulsified oils and for sediment and phase separation. Use strong transmitted light for observations of such deficiencies. A light fitted with a shield having a narrow slot (about 1/4 by 4 inches) for viewing is suitable for this purpose.

4.4.8 Storage stability.

4.4.8.1 Type L. Shake the 1-pint composite sample thoroughly. Fill 100 ml. glass-stoppered,

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amber, glass cylinder graduated in 1.0 ml. units exactly to the 100 ml. with disinfectant concentrate. Store the original container and the glass cylinder at 25° ± 3° C. (77° ± 5.4° F.) on a level shelf in an area protected from light for 1 year. After this period, test the contents of the original container for antimicrobial potency specified in 4.4.2. In addition, remove the glass cylinder from the dark area carefully without agitation and read the volume of sediment.

4.4.8.2 Type S. Store the 1-pound composite sample at 25° ± 3° C. (77° ± 5.4° F.) in an area protected from light for 1 year. After this period, test for antimicrobial potency specified in 4.4.2.

4.4.8.3 Certificate of storage stability. A certificate of storage stability shall be furnished the contracting officer or his representative certifying that the end item meets this requirement (see 3.1.1).

4.4.9 Corrosive effect.

4.4.9.1 Stainless steel. Immerse two scalpel blades, conforming to GG-H-80, in 100 ml. of a 1:50 dilution of the disinfectant concentrate for a period of 16 hours at 25° C. The addition of corrosion inhibitors, such as sodium nitrate, shall be permitted provided adequate directions for such use appear on the label furnished in accordance with 3.11. Remove, rinse the blades in distilled water, and dry. Examine for signs of corrosion.

4.4.9.2 Galvanized iron. Cut six 2 by 2 inch test pieces from 18 gauge galvanized iron sheet and wash them in carbon tetrachloride. Immediately before the test, wipe the test pieces with a clean cloth moistened with carbon tetrachloride, then dry at 105° C. for thirty minutes and cool in a desiccator. Weigh each test piece to the nearest 0.1 milligram to obtain the initial weight. Immerse the six test pieces in three separate beakers (two to a beaker), each containing 500 ml. of 1:50 dilution of the disinfectant. The test pieces shall not touch each other and shall be completely immersed in the test solutions. The beakers shall be covered with a watch glass and immersed to the solution level in a constant-temperature water bath maintained at 35° ± 1° C. After a 24-hour contact period, the test strips shall be removed from the disinfectant, thoroughly rinsed in distilled water, wiped with a clean towel, and dried in an oven at 105° C., cooled in a desiccator, and re-weighed. The weight change shall be computed by taking the arithmetic mean of the six test pieces.

4.4.9.2.1 Calculation.

$$\text{Weight change per strip} = (\text{Initial weight}) - (\text{Final weight})$$

$$\text{Average weight change} = \frac{\text{Total weight change}}{\text{Number of test pieces}}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type L. The type L concentrate shall be furnished in 1-quart, 1-gallon, or 5-gallon quantities as specified (see 6.2).

5.1.1.1.1 One-quart bottles. Unless otherwise specified (see 6.2), the one-quart bottles shall conform to PPP-C-186, group A, class 1, style 1, grade 3, amber-colored glass, closure A, outer and inner seals A and B. When specified (see 6.2), the one-quart bottle shall be made from high density polyethylene and conform to PPP-C-186, group A, class 2, style 1, grade 3, amber colored, closure A, outer and inner seals A and B. Twelve bottles of concentrate shall then be packaged in accordance with procedure Code No. 7 of PPP-C-186, except that partitions and liners for boxes with plastic bottles, shall be the full height of the bottle plus one-fourth of an inch.

5.1.1.1.2 One-gallon bottles. Unless otherwise specified (see 6.2), the one-gallon bottles shall conform to PPP-C-186, group A, class 1, style 1, grade 3, amber-colored glass, closure A, outer seal A. When specified (see 6.2), the one-gallon bottle shall be made from high density polyethylene and conform to PPP-C-186, group A, class 2, style 1, grade 3, amber-color, closure A, outer seal A. Each glass bottle of concentrate shall be packaged in accordance with procedure Code No. 8 of PPP-C-186.

5.1.1.1.3 Five-gallon pails. Unless otherwise specified (see 6.2), the five-gallon pails shall conform to PPP-P-704, type I, class 3, with interior coating. Each pail shall be provided with a screw-cap closure, push-pull self-venting spout, and a tamperproof seal. When specified (see 6.2), the five-gallon container shall be a polyethylene tight head pail made from material conforming to MIL-B-26701 and provided with a flexible spout and vent. The minimum side wall shall be

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0.0625-inch thick, and the minimum weight shall be 3-1/2 pounds.

5.1.1.2 Type S. The type S concentrate shall be furnished in 5-pound square or oblong, metal cans. The 5-pound metal can shall conform to PPP-C-96, type V, class 1, exterior coating plan B.

5.1.2 Level C. The disinfectant concentrate shall be packaged to afford adequate protection against deterioration or damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Disinfectant concentrate, packaged as specified in 5.1, shall be packed: twelve 1-quart bottles, six 1-gallon bottles, one 5-gallon polyethylene pail, or eight 5-pound cans in a snug-fitting fiberboard shipping container conforming to PPP-B-636, grade V2s. The shipping container containing the six one-gallon polyethylene bottles shall be provided with two one-piece, taped liners placed in such a manner that each liner shall encase three bottles. The liners shall be the full height of the bottles plus one-fourth of an inch and shall be fabricated from fiberboard conforming to PPP-F-320, grade V3c. The 5-gallon steel pails of concentrate shall not require any overpacking.

5.2.2 Level B. Unless otherwise specified (see 5.2.2.1), disinfectant concentrate, packaged as specified in 5.1, shall be packed: twelve 1-quart bottles, six 1-gallon bottles, one 5-gallon polyethylene pail, or eight 5-pound cans, in a snug-fitting fiberboard shipping container conforming to PPP-B-636, type CF, variety SW, or type SF, class domestic. The shipping container containing the six 1-gallon polyethylene bottles shall be provided with two liners as specified in 5.2.1, except that the liner shall be fabricated from the same material as the shipping container. Each shipping containrner shall be closed in accordance with method II of the appendix of PPP-B-636. The 5-gallon steel pails of concentrate shall not require any overpacking.

5.2.2.1 When specified (see 6.2), the shipping container shall conform to PPP-B-636, grade V3c or V3s.

5.2.3 Level C. Disinfectant concentrate, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The disinfectant concentrate covered by this specification is intended to be used as follows:

(a) To be used for general disinfection, as in floors, walls, ceilings, urinals, garbage pails, washstands, and other objects.
(b) To be used for emergency disinfection of surgical instruments when sterilization by autoclave is not possible.
(c) Not intended to be used:
  (1) For disinfection where a high concentration of organic material is present.
  (2) As a substitute for disinfection, latrine bucket (MIL-D-51061) or disinfectant, chlorine, food service (MIL-D-11309).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type concentrate desired (see 1.2).
(c) For type S concentrate, the multiple of standard strength should be stated (see 4.4.1).
(d) Quantity of concentrate desired (see 5.1.1.1).
(e) When one-quart or one-gallon polyethylene bottles are required (see 5.1.1.1.1 or 5.1.1.1.2).
(f) Whether weather-resistant class fiberboard shipping containers are required for

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level B shipment (see 5.2.2.1).
(g) Selection of the applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Purchase. Type L disinfectant concentrate is purchased by volume, the unit being a U. S. quart or U. S. gallon of 20° C. (68° F.). Type S disinfectant concentrate is purchased by weight, the unit being an avoirdupois pound.

MILITARY CUSTODIANS:

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Navy - SH
Air Force - 68

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User activities:

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Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Disinfectant, Germicidal and Fungicidal, Concentrate- O-D-406a

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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